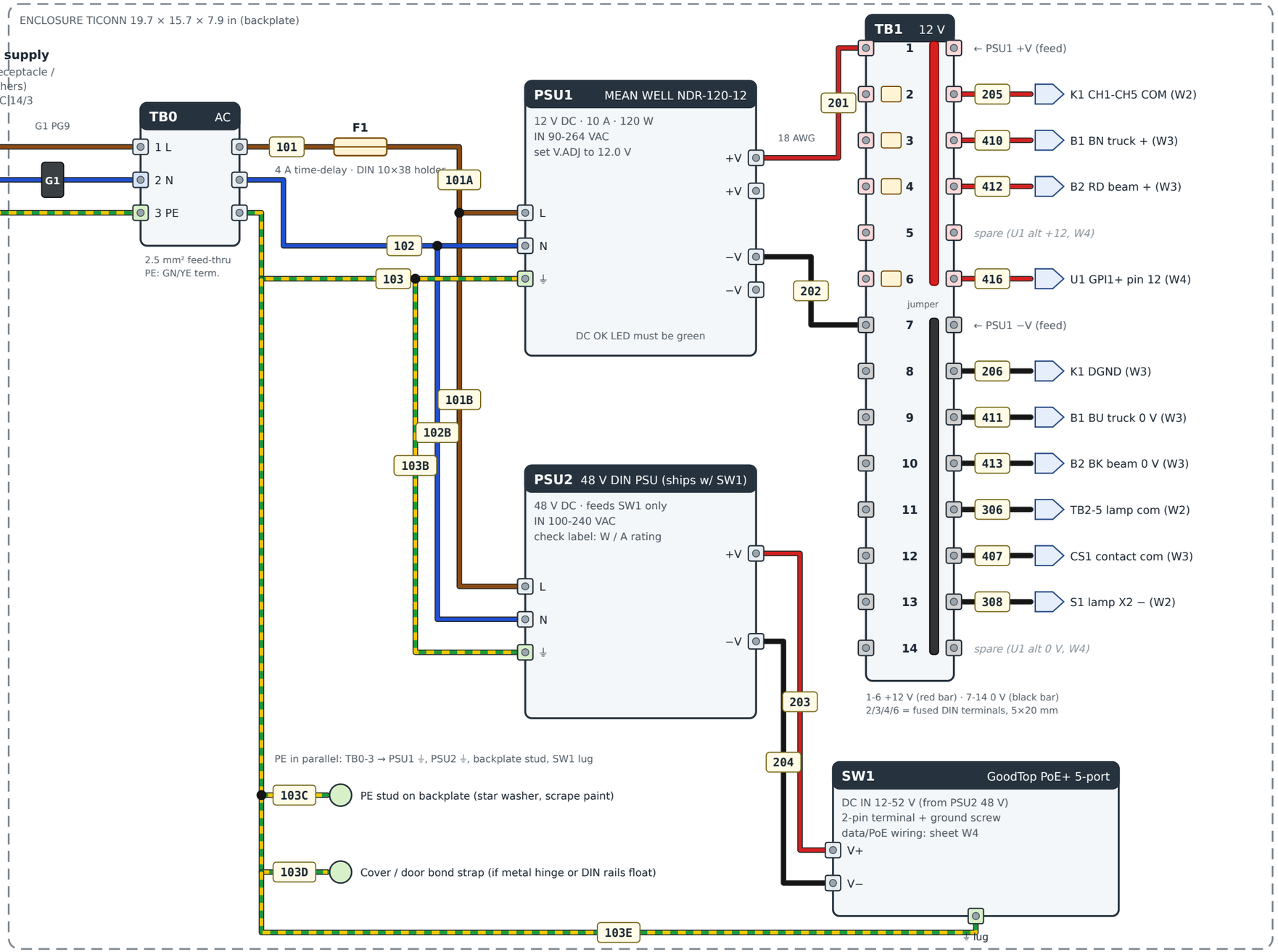
# W1 · Power distribution

120 VAC in through G1, one fuse, two DIN supplies. K1 is PoE-powered: leave its 7-36 V input EMPTY.




## Power budget (per box)

**PSU1 12 V (120 W available)**
2 stack lights, 1 lamp + buzzer each ≈ 6 W
S1 pilot lamp (12 V LED) ≈ 0.3 W
B1 laser + B2 beam ≈ 2 W
total ≈ 8-11 W (<10 %)
U1 reader if moved to PSU1 (W4 option) +36 W max
**PSU2 48 V → SW1**
SW1 + PoE: SPL1/U1 ≤ 25.5 W, K1 ≈ 3-5 W → check label
**F1 (120 VAC)**
4 A time-delay (Class CC/midget 10×38). NDR-120 inrush
≈ 30 A cold start: do NOT use fast-acting.
If PSU2 label > 1.5 A in, go 5 A.
**F2-F4 (12 V branch, 5×20 mm in fused terminals)**
F2 2 A K1 relay COM bus · F3/F4 0.5 A sensors · F5 0.5 A U1 GPI1+
Protects 18-22 AWG field cable from the 10 A supply.

## Power-up check (before K1/U1 connected)

1. Meter TB0-1 → TB0-2 ≈ 120 VAC, TB0-1 → TB0-3 ≈ 120 VAC.
2. PE continuity TB0-3 ↔ backplate stud ↔ PSU cases < 1 Ω.
3. Close F1. PSU1 DC OK lit; TB1-1 → TB1-7 = 12.0 V (trim V.ADJ).
4. TB1-2/3/4/6 → TB1-7 = 12 V (fuses in).
5. PSU2 +V → −V = 48 V; SW1 power LED on.
6. K1 7-36 V terminal: nothing landed. PoE only.
7. TB1 0 V is NOT bonded to PE (floating DC system).

## Wire & termination spec

AC (101-103): 18 AWG MTW/UL1015 600 V, BN L · BU N · GN/YE PE
DC 12 V / 48 V: 18 AWG UL1007, RD + · BK 0 V
All stranded ends: insulated ferrules (18 AWG = 1.0 mm²)
Two wires in one terminal: twin ferrule (PSU1 L, N, ⏚)
Torque: PSU 4.4 lb·in · DIN terminals 5 lb·in (check part)
Label both ends of every wire with its number

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

| | | |
|---|---|---|
| **RFID PANEL BOX · RPB-001 · WIRING DIAGRAM**<br>Gielow Pickles · G5 Croswell · one box per dock door | REV<br>**9 (proposed)** | SHEET<br>**1 of 5** |
| **W1 AC + DC POWER DISTRIBUTION**<br>G1 → TB0 → F1 → PSU1 / PSU2 → TB1 12 V bus, SW1 48 V | DATE<br>Oct 28, 2026 | CHECKED<br>_____ |

# W2 · K1 relay outputs → stack lights, buzzer, S1 lamp

Relay commons are jumpered to +12 V (fused F2). Relays switch +12 V out on NO. Tower common returns to 0 V.

TB1-2 +12 V via F2 2 A (W1)

**K1** Waveshare ESP32-S3-POE-ETH-8DI-8RO

relay section · 10 A 250 VAC / 30 VDC
TCA9554 @0x20, bit n = CHn+1

| | | |
|---|---|---|
| **CH1** | RED lamp | NO · C · NC |
| **CH2** | AMBER lamp | NO · C · NC |
| **CH3** | GREEN lamp | NO · C · NC |
| **CH4** | BUZZER | NO · C · NC |
| **CH5** | S1 pilot lamp | NO · C · NC |
| **CH6** | spare | NO · C · NC |
| **CH7** | spare | NO · C · NC |
| **CH8** | spare | NO · C · NC |

205 jumpers
C1→C2→C3→C4→C5
18 AWG RD, ferruled

205 · 301 · 205 · 302 · 303 · 304 · 305 · 306

**TB2** LAMPS

4-level / quattro
DIN terminals

1 RED · 2 AMBER · 3 GREEN · 4 BUZZ · 5 COM 0V

306

TB1-11 0 V (W1)

Panel wiring K1→TB2: 18 AWG, color as shown.

**S1** LOAD COMPLETE lamp (green)

22 mm illuminated PB, 12 V LED
X1 + / X2 − contacts: W3

G7 · X1 + · X2 − · 308

TB1-13 0 V (W1) · in cable W-CS (6-cond), see W3

**HL1** 12 V LED tower R/A/G + buzzer

5-wire lead · verify colors on arrival

301 · G5 · 306

RED · AMBER · GREEN · BUZZER · common

Cable W-HL1: tower lead or 18/5 SJOOW to G5

**HL2** 12 V LED tower R/A/G + buzzer

5-wire lead · verify colors on arrival

301 · G6 · 306

RED · AMBER · GREEN · BUZZER · common

Cable W-HL2: tower lead or 18/5 SJOOW to G6
Lead too short → splice in IP67 JB at mast, keep colors

Terminal order on the real board: VERIFY (pluggable 3-pin per CH, silkscreen NO / C / NC). Relay LEDs mirror coil state.

### Stack-light notes

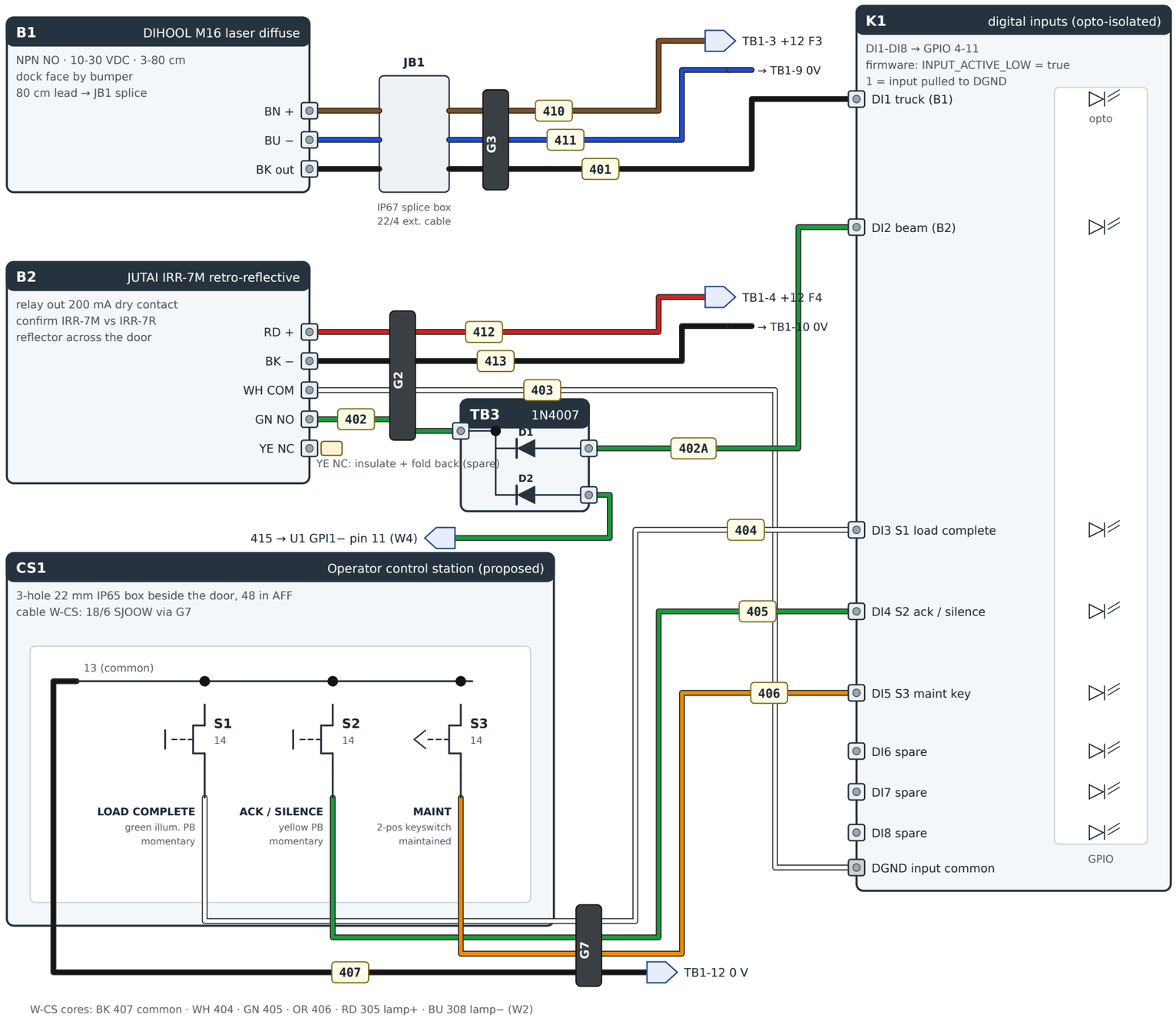
Lamp truth: firmware sets exactly ONE of CH1/CH2/CH3. CH4 = buzzer, CH5 = S1 lamp.
HL1 + HL2 in parallel at TB2 (one wire in, two out per level). 4-level terminals,
or 2 stacked 2-level terminals with a bridge. Each tower ≈ 3 W → F2 2 A is plenty.
Tower common polarity: meter it. If the common must be +12 V (common-anode tower):
move 205 to TB1 0 V and 306 to TB1 +12 V. Nothing else changes.
Typical lead colors: RD red · YE amber · GN green · WH or BN buzzer · BK common.
Relay NC contacts unused: leave empty. Don't wire anything to NC on CH1-CH3.

| | | |
|---|---|---|
| **RFID PANEL BOX · RPB-001 · WIRING DIAGRAM**<br>Gielow Pickles · G5 Croswell · one box per dock door | REV<br>**9 (proposed)** | SHEET<br>**2 of 5** |
| **W2 K1 RELAY OUTPUTS**<br>CH1-CH4 → TB2 → HL1 + HL2 stack lights · CH5 → S1 lamp · CH6-8 spare | DATE<br>Oct 28, 2026 | CHECKED<br>_____ |

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

# W3 · K1 digital inputs → sensors + operator controls

Opto inputs switch when DIn is pulled to DGND (NPN sink / dry contact). DGND bonded to TB1 0 V.




W-CS cores: BK 407 common · WH 404 · GN 405 · OR 406 · RD 305 lamp+ · BU 308 lamp− (W2)

## Input map

| Input | Device | Closed / 1 means | Wire |
|---|---|---|---|
| **DI1** | B1 BK (NPN NO) | truck present | 401 |
| **DI2** | B2 NO via D1 / COM | beam broken* | 402A/403 |
| **DI3** | S1 13-14 | load complete pressed | 404 |
| **DI4** | S2 13-14 | ack / silence pressed | 405 |
| **DI5** | S3 13-14 | maintenance ON | 406 |
| **DI6-8** | — | spare (dry contact to DGND) | — |
| **DGND** | TB1-8 | bonded to 0 V | 206 |
| **U1 GPI1** | B2 NO via D2 | reader trigger (W4) | 415/416 |

### Why it fails safe

B1 is NPN normally-open: cut cable / dead sensor reads 'no truck' → server shows RED.
*B2 relay: pick NO or NC on the bench so 'beam broken' = 1. With NO + light-on,
a cut cable reads 'beam clear' → reader stays idle (no false green).
S3 maint is maintained; key removable only in NORMAL.
These inputs are NOT a safety circuit. Don't use them for machine guarding / E-stop.

### Bench test (README) — do this before mounting

1. DGND→TB1 0 V landed (206). Hand in front of B1 → serial 'DI1 -> 1'.
2. Block B2 → 'DI2' toggles. Move GN→YE (NC) if it reads backwards.
3. Press S1, S2; turn S3 → DI3/DI4/DI5 → 1, inputs JSON release/ack/maint true.
4. If every DI reads inverted, set INPUT_ACTIVE_LOW false in config.h.
5. Block B2 with reader in trigger mode → U1 starts reading (W4).
6. If a dry contact to DGND does NOT switch the opto, the board needs a
sourced input: see option B below.

### Option B — if K1 DI turns out to need an external voltage

Waveshare DI wiring is still UNVERIFIED (README 'known unknowns').
If DIn needs +V to switch (opto anode at DIn, cathode at DGND):
· DGND stays on 0 V (206).
· B2 WH COM and S1-S3 common (407) move from 0 V to +12 V (TB1-5/6).
· B1 NPN can't source: swap to a PNP sensor (BK = +12 when target seen),
e.g. the PNP Heschen from PO 1811, or add a 12 V interposing relay.
Mark the change on this sheet; firmware is unchanged.

### Why TB3 diodes (one beam contact, two inputs)

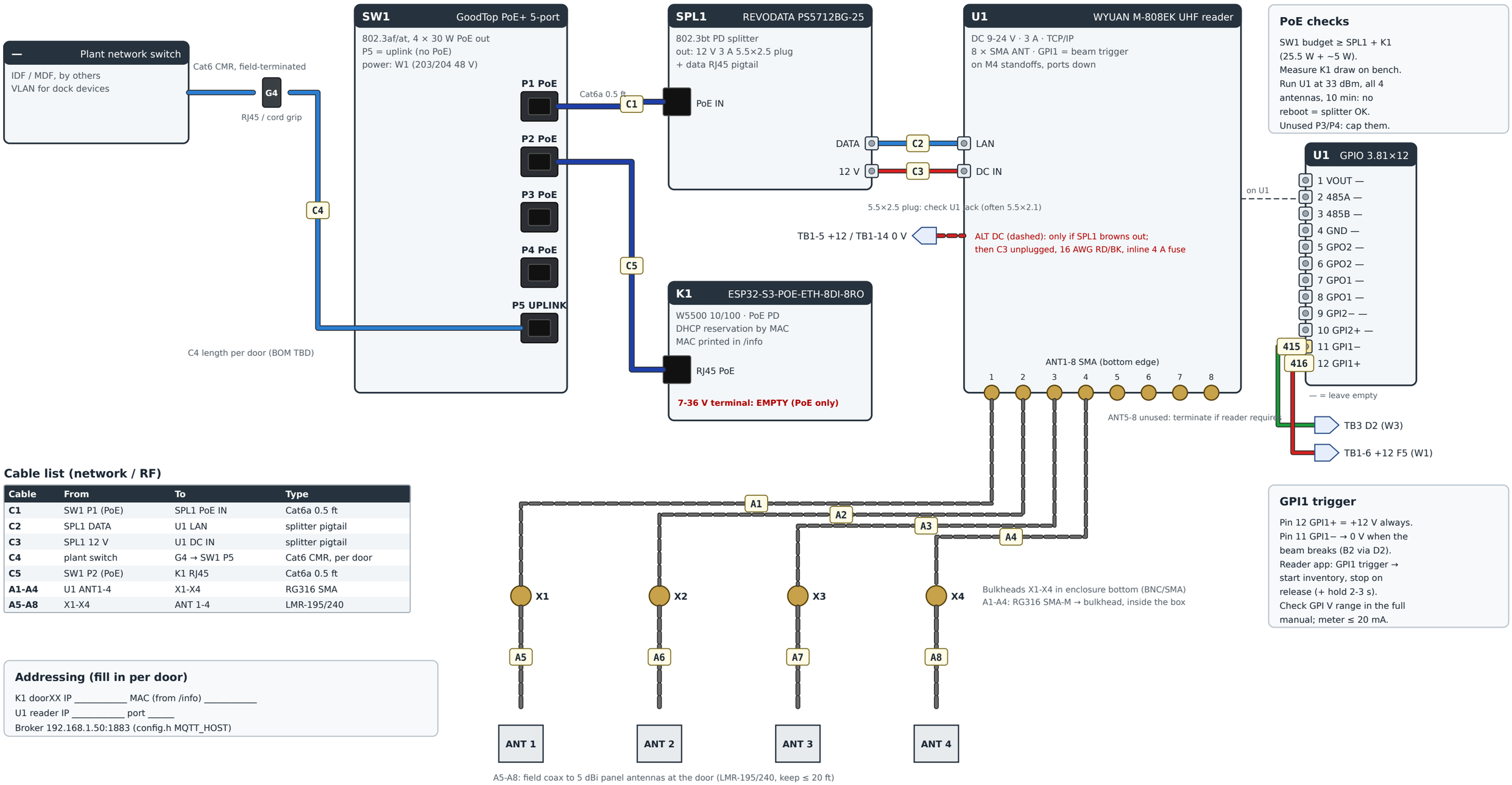
B2 has ONE relay contact. Tied straight to DI2 and GPI1−, the open contact leaves
+12 → GPI1 LED → K1 DI2 LED → 0 V in series: both can false-trigger. D1/D2 block it.
Contact closed: both cathodes at 0 V → DI2 = 1 and GPI1 on (≈ 0.7 V drop, fine).
B2 22/5 lead · B1 22/4 ext · CS1 18/6 SJOOW. Keep away from AC + coax.

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

| | | |
|---|---|---|
| **RFID PANEL BOX · RPB-001 · WIRING DIAGRAM**<br>Gielow Pickles · G5 Croswell · one box per dock door | REV<br>**9 (proposed)** | SHEET<br>**3 of 5** |
| **W3 K1 DIGITAL INPUTS**<br>B1 truck → DI1 · B2 beam → DI2 · S1/S2/S3 control station → DI3-DI5 | DATE<br>Oct 28, 2026 | CHECKED<br>_____ |

# W4 · Network, PoE and antenna wiring

Everything is copper Ethernet + PoE inside the box. Plant network enters on C4 through G4.




### PoE checks

SW1 budget ≥ SPL1 + K1 (25.5 W + ~5 W).
Measure K1 draw on bench.
Run U1 at 33 dBm, all 4 antennas, 10 min: no reboot = splitter OK.
Unused P3/P4: cap them.

### Cable list (network / RF)

| Cable | From | To | Type |
|---|---|---|---|
| **C1** | SW1 P1 (PoE) | SPL1 PoE IN | Cat6a 0.5 ft |
| **C2** | SPL1 DATA | U1 LAN | splitter pigtail |
| **C3** | SPL1 12 V | U1 DC IN | splitter pigtail |
| **C4** | plant switch | G4 → SW1 P5 | Cat6 CMR, per door |
| **C5** | SW1 P2 (PoE) | K1 RJ45 | Cat6a 0.5 ft |
| **A1-A4** | U1 ANT1-4 | X1-X4 | RG316 SMA |
| **A5-A8** | X1-X4 | ANT 1-4 | LMR-195/240 |

### GPI1 trigger

Pin 12 GPI1+ = +12 V always.
Pin 11 GPI1− → 0 V when the beam breaks (B2 via D2).
Reader app: GPI1 trigger → start inventory, stop on release (+ hold 2-3 s).
Check GPI V range in the full manual; meter ≤ 20 mA.

### Addressing (fill in per door)

K1 doorXX IP _____ MAC (from /info) _____
U1 reader IP _____ port _____
Broker 192.168.1.50:1883 (config.h MQTT_HOST)

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

| | | |
|---|---|---|
| **RFID PANEL BOX · RPB-001 · WIRING DIAGRAM**<br>Gielow Pickles · G5 Croswell · one box per dock door | REV<br>**9 (proposed)** | SHEET<br>**4 of 5** |
| **W4 NETWORK · PoE · RF**<br>plant switch → G4 → SW1 → SPL1 → U1 · SW1 → K1 · U1 → 4 antennas | DATE<br>Oct 28, 2026 | CHECKED<br>_____ |

# W5 · Wire, terminal and gland schedules

Build from this list. Check each row off as you land it, then ring it out end to end.

## Wire schedule — power + outputs

| Wire | From | To | Color | AWG | Sht |
|---|---|---|---|---|---|
| **101** | G1 field L | TB0-1 | BN | 18 | W1 ☐ |
| **101** | TB0-1 | F1 in | BN | 18 | W1 ☐ |
| **101A** | F1 out | PSU1 L | BN | 18 | W1 ☐ |
| **101B** | PSU1 L | PSU2 L | BN | 18 | W1 ☐ |
| **102** | TB0-2 | PSU1 N | BU | 18 | W1 ☐ |
| **102B** | PSU1 N | PSU2 N | BU | 18 | W1 ☐ |
| **103** | TB0-3 | PSU1 ⏚ | GN/YE | 18 | W1 ☐ |
| **103B** | PSU1 ⏚ | PSU2 ⏚ | GN/YE | 18 | W1 ☐ |
| **103C** | TB0-3 | backplate PE stud | GN/YE | 14 | W1 ☐ |
| **103D** | PE stud | cover bond | GN/YE | 14 | W1 ☐ |
| **103E** | PE stud | SW1 ground lug | GN/YE | 18 | W1 ☐ |
| **201** | PSU1 +V | TB1-1 | RD | 18 | W1 ☐ |
| **202** | PSU1 −V | TB1-7 | BK | 18 | W1 ☐ |
| **203** | PSU2 +V | SW1 V+ | RD | 18 | W1 ☐ |
| **204** | PSU2 −V | SW1 V− | BK | 18 | W1 ☐ |
| **205** | TB1-2 (F2) | K1 CH1 C | RD | 18 | W2 ☐ |
| **205** | K1 CH1 C | CH2 C, CH3 C, CH4 C, CH5 C | RD | 18 | W2 ☐ |
| **206** | TB1-8 | K1 DGND | BK | 18 | W3 ☐ |
| **301** | K1 CH1 NO | TB2-1 → HL1/HL2 red | RD | 18 | W2 ☐ |
| **302** | K1 CH2 NO | TB2-2 → HL1/HL2 amber | OR/YE | 18 | W2 ☐ |
| **303** | K1 CH3 NO | TB2-3 → HL1/HL2 green | GN | 18 | W2 ☐ |
| **304** | K1 CH4 NO | TB2-4 → HL1/HL2 buzzer | WH | 18 | W2 ☐ |
| **305** | K1 CH5 NO | G7 → S1 X1 (lamp +) | RD | 18 | W2 ☐ |
| **306** | TB1-11 | TB2-5 → HL1/HL2 common | BK | 18 | W2 ☐ |
| **308** | TB1-13 | G7 → S1 X2 (lamp −) | BU | 18 | W2 ☐ |

## Wire schedule — inputs + network

| Wire | From | To | Color | AWG | Sht |
|---|---|---|---|---|---|
| **401** | B1 BK (JB1) | G3 → K1 DI1 | BK | 22 | W3 ☐ |
| **402** | B2 GN (NO) | G2 → TB3 node (D1/D2 K) | GN | 22 | W3 ☐ |
| **402A** | TB3 D1 anode | K1 DI2 | GN | 18 | W3 ☐ |
| **415** | TB3 D2 anode | U1 GPIO 11 GPI1− | GN | 22 | W3/W ☐ |
| **416** | TB1-6 (F5) | U1 GPIO 12 GPI1+ | RD | 22 | W1/W ☐ |
| **403** | B2 WH (COM) | G2 → K1 DGND | WH | 22 | W3 ☐ |
| **404** | S1 14 | G7 → K1 DI3 | WH | 18 | W3 ☐ |
| **405** | S2 14 | G7 → K1 DI4 | GN | 18 | W3 ☐ |
| **406** | S3 14 | G7 → K1 DI5 | OR | 18 | W3 ☐ |
| **407** | TB1-12 | G7 → S1/S2/S3 13 (jumpered) | BK | 18 | W3 ☐ |
| **410** | TB1-3 (F3) | G3 → JB1 → B1 BN | BN/RD | 22 | W3 ☐ |
| **411** | TB1-9 | G3 → JB1 → B1 BU | BU/BK | 22 | W3 ☐ |
| **412** | TB1-4 (F4) | G2 → B2 RD | RD | 22 | W3 ☐ |
| **413** | TB1-10 | G2 → B2 BK | BK | 22 | W3 ☐ |
| **—** | B2 YE (NC) | insulated, spare | YE | 22 | W3 ☐ |
| **C1** | SW1 P1 | SPL1 PoE IN | Cat6a | — | W4 ☐ |
| **C2** | SPL1 data | U1 LAN | pigtail | — | W4 ☐ |
| **C3** | SPL1 12 V | U1 DC IN | pigtail | — | W4 ☐ |
| **C4** | plant switch | G4 → SW1 P5 | Cat6 | — | W4 ☐ |
| **C5** | SW1 P2 | K1 RJ45 | Cat6a | — | W4 ☐ |
| **A1-4** | U1 ANT1-4 | X1-X4 | RG316 | — | W4 ☐ |
| **A5-8** | X1-X4 | ANT 1-4 | LMR | — | W4 ☐ |

## Terminal strips

| Terminal | Function | Wires |
|---|---|---|
| **TB0-1** | L in (G1) | 101 → F1 |
| **TB0-2** | N in (G1) | 102 → PSU1 |
| **TB0-3** | PE in (G1) | 103 … |
| **TB1-1** | +12 feed | 201 PSU1 +V |
| **TB1-2** | +12 F2 2 A | 205 K1 COMs |
| **TB1-3** | +12 F3 ½ A | 410 B1 |
| **TB1-4** | +12 F4 ½ A | 412 B2 |
| **TB1-5** | +12 | spare / U1 alt |
| **TB1-6** | +12 F5 ½ A | 416 U1 GPI1+ |
| **TB1-7** | 0 V feed | 202 PSU1 −V |
| **TB1-8** | 0 V | 206 K1 DGND |
| **TB1-9** | 0 V | 411 B1 |
| **TB1-10** | 0 V | 413 B2 |
| **TB1-11** | 0 V | 306 TB2-5 |
| **TB1-12** | 0 V | 407 CS1 common |
| **TB1-13** | 0 V | 308 S1 lamp − |
| **TB1-14** | 0 V | spare / U1 alt |
| **TB2-1** | red | 301 in / HL1 + HL2 |
| **TB2-2** | amber | 302 in / HL1 + HL2 |
| **TB2-3** | green | 303 in / HL1 + HL2 |
| **TB2-4** | buzzer | 304 in / HL1 + HL2 |
| **TB2-5** | lamp com | 306 in / HL1 + HL2 |
| **TB3-D1** | diode → DI2 | 402 K / 402A A |
| **TB3-D2** | diode → GPI1− | 402 K / 415 A |

## Glands (bottom face)

| Gland | For | Cable | Size |
|---|---|---|---|
| **G1** | AC in 120 V | 18/3 or 14/3 | PG9 |
| **G2** | B2 beam | 22/5 | PG7 |
| **G3** | B1 truck (JB1) | 22/4 | PG7 |
| **G4** | C4 uplink | Cat6 | PG11 split |
| **G5** | HL1 tower | 18/5 | PG9 |
| **G6** | HL2 tower NEW | 18/5 | PG9 |
| **G7** | CS1 station NEW | 18/6 | PG9 |
| **X1-X4** | antenna bulkheads | coax | BNC/SMA |

### Rev 9 changes vs rev 8 (proposed — mark up the DXF when approved)

1. TB0 AC entry terminals (L / N / PE) and a backplate PE stud; PE to PSU1, PSU2, SW1, cover.
2. TB1 grows to 14 positions (6 × +12 V, 8 × 0 V); TB1-2/3/4 become fused terminals F2/F3/F4.
3. TB2 lamp marshalling (5 × 4-level terminals) so HL1 and HL2 each land on their own clamp.
4. HL2 gets its own gland G6 (rev 8 shared G5 between two cables = not IP67).
5. Operator control station CS1 (S1 load complete, S2 ack/silence, S3 maint key) on K1 DI3-DI5, S1 lamp on K1 CH5. New gland G7, cable W-CS 18/6. Wires 305, 308, 404-407.
6. 48 V return gets its own number 204 (rev 8 used 203 for both legs). Sensor supplies 410-413.
7. Firmware 1.1.0: buzzer/pilot topics, blink states, server-presence fail-safe, /info fw + MAC.
8. B2 beam also triggers U1 GPI1 (pin 12 +12 V F5, pin 11 sinks via TB3 diode D2). Wires 402A, 415, 416.

### Added parts for rev 9 (per box, not yet priced)

TB0: 2 × feed-thru + 1 × PE ground terminal, 2.5 mm² · TB1: 8 more terminals + 3 fused terminal holders
TB2: 5 × 4-level (quattro) terminals + end plate · jumper bars · PE stud kit · TB3: 2 × 1N4007 diode terminals
CS1: 3-hole 22 mm IP65 station, green illuminated 12 V PB, yellow PB, 2-pos keyswitch, legend plates
Cable: 18/6 SJOOW for CS1 · 18/5 for HL2 · 2 × PG9 glands (G6, G7) · fuses 2 A + 2 × 0.5 A (5×20)

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

| | | |
|---|---|---|
| **RFID PANEL BOX · RPB-001 · WIRING DIAGRAM**<br>Gielow Pickles · G5 Croswell · one box per dock door | REV<br>**9 (proposed)** | SHEET<br>**5 of 5** |
| **W5 WIRE · TERMINAL · GLAND SCHEDULES**<br>every conductor, from → to, with color, gauge and sheet | DATE<br>Oct 28, 2026 | CHECKED<br>_____ |